Theorem. 1.4.4

Convergent sequences are bounded.

Proof. Let $\{x_n\}$ be a convergent sequence, i.e. $x_n \longrightarrow L \in \mathbb{R}$, as $n \longrightarrow \infty$, for some limit L.

Let $\varepsilon = 1$, since $\{x_n\}$ is convergent there exists N such that

$$n \geq N \Rightarrow |x_n - L| < 1$$

i.e. $n \geq N \Rightarrow L - 1 < x_n < L + 1$

Let

$$M = \max\left(|x_1|, |x_2|, \ldots, |x_{N-1}|, |L| + 1 \right)$$

Now we can say for all $n \geq 1$

$$|x_n| \leq M$$

So the sequence is bounded.

~~Theorem 1.4.8~~

Consider the application of the theorem in Example 1.4.10

$$= \dots = \frac{\lim (2n^2 - n)}{\lim (-3n^2 + 2)} \quad ???$$

WRONG as these limits do not exist.

Proof of linearity of convergence

Assume $\alpha, \beta \neq 0$. ~~Kete~~ Assume $a_n \to a$, $b_n \to b$ as $n \to \infty$.

Claim: $\lim_{n\to\infty} (\alpha a_n + \beta b_n) = \alpha a + \beta b$.

Investigate the final equality.

$$|\alpha a_n + \beta b_n - (\alpha a + \beta b)| < \varepsilon$$

$$\Leftrightarrow |\alpha(a_n - a) + \beta(b_n - b)| < \varepsilon$$

Notice by the triangle inequality
($|x+y| \leq |x| + |y|$).

want $< \varepsilon$

$$|\alpha(a_n - a) + \beta(b_n - b)|$$

$$\leq |\alpha(a_n - a)| + |\beta(b_n - b)|$$

$$= |\alpha||a_n - a| + |\beta||b_n - b|, \quad \text{using } |xy| = |x||y|$$

$$< \varepsilon/2 + \varepsilon/2.$$

want $< \varepsilon$

Let's make $|a_n - a| < \frac{\varepsilon}{|\alpha| 2}$.

" " $|b_n - b| < \frac{\varepsilon}{|\beta| 2}$.

Here's the proof.

Let $\varepsilon > 0$ be given.

By convergence there exists points N_1, N_2

such that

$n \geq N_1 \qquad |a_n - a| < \frac{\varepsilon}{2|\alpha|}$

and

$n \geq N_2 \qquad |b_n - b| < \frac{\varepsilon}{2|\beta|}$

Now let $N = \max(N_1, N_2)$

so for $n \geq N$ both inequalities will hold

Therefore for all $n \geq N$

$$|\alpha a_n + \beta b_n - (\alpha a + \beta b)|$$

$$\leq |\alpha|\,|a_n - a| + |\beta|\,|b_n - b|, \text{ from above.}$$

$$< |\alpha| \frac{\varepsilon}{2|\alpha|} + |\beta| \frac{\varepsilon}{2|\beta|}$$

$$= \varepsilon.$$

This proves the definition for

$\alpha a_n + \beta b_n \to \alpha a + \beta b$ as $n \to \infty$.

(continue at 12:08).

Product case and quotient case are proved in a similar manner.

Notes $|x_n - L| < \varepsilon$

$|a_n - a|$

$|b_n - b|$

$$|a_n b_n - ab|$$

$$= |(a_n - a)(b_n - b) - 2ab + a_n b + ab_n|$$

$$= |(a_n - a)(b_n - b) + b(a_n - a) + a(b_n - b)|$$

$$\leq |a_n - a||b_n - b| + |b||a_n - a| + |a||b_n - b|$$

$|a_n - a| < \sqrt{\varepsilon/3}$ $|b_n - b| < \sqrt{\varepsilon/3}$ $|a_n - a| < \frac{\varepsilon}{3|b|}$ $|b_n - b| < \frac{\varepsilon}{3|a|}$

$$< \varepsilon/3 + \varepsilon/3 + \varepsilon/3$$

want

$$< \varepsilon$$

Reading: Study carefully all parts of proof of algebra of limits theorem.

Examples concerning Monotone Convergence Theorem (proofs by induction).

Q9 from Tutorial sheet.

$a_0 = 1$. Claim for all $n \geq 0$ $a_n \geq 1$.

ie. claim: 1 is a lower bound.

Base case: $a_0 = 1 \geq 1$ True at beginning

Assume that $a_k \geq 1$ for some $k \geq 0$

$$\Rightarrow 0 < \frac{1}{a_k} \leq 1$$

$$\Rightarrow 3 - \frac{1}{a_k} \geq 2.$$

ie. $a_{k+1} \geq 2 \geq 1$

By the principle of induction $a_n \geq 1$ for all $n \geq 0$. $\quad 1 = a_0 < 3$.

And then since $a_n \geq 1$

$$a_{n+1} = 3 - \frac{1}{a_n} \leq 3.$$

So 3 is an upper bound for $\{a_n\}$

So $\{a_n\}$ is bounded above and below.

Let's also use induction to prove sequence always increase.

$$a_0 = 1, \quad a_1 = 3 - \frac{1}{a_0} = 2.$$

"Base case"

So $\quad a_0 < a_1$

Assume that $a_k < a_{k+1}$ for some $k \geq 0$

$$0 < a_k < a_{k+1}$$

$$\Rightarrow 0 < \frac{1}{a_{k+1}} < \frac{1}{a_k}$$

$$\Rightarrow -\frac{1}{a_{k+1}} > -\frac{1}{a_k}$$

$$\Rightarrow \quad 3 - \frac{1}{a_{k+1}} > 3 - \frac{1}{a_k}.$$

$$\Rightarrow \quad \boxed{a_{k+2} > a_{k+1}}$$

So by induction, $\{a_n\}$ is increasing.

So by the Monotone Convergence Theorem $\{a_n\}$ is convergent.

Let's say $a_n \to a \in \mathbb{R}$, as $n \to \infty$.

Let's discover the value of a by "taking the limit" of the recurrence.

for all $n \geq 0$. (We know $a \neq 0$)

$$a_n = 3 - \frac{1}{a_{n-1}}$$

$$\Rightarrow \lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(3 - \frac{1}{a_{n-1}}\right)$$

$$\Rightarrow \quad a = 3 - \frac{1}{\lim_{n\to\infty} a_{n-1}}, \quad \text{by algebra of limits theorem}$$

$$\Rightarrow \quad a = 3 - \frac{1}{a}$$

$$\Rightarrow \quad a^2 = 3a - 1$$

$$\Rightarrow \quad a^2 - 3a + 1 = 0$$

$$\Rightarrow \quad a = \frac{3 \pm \sqrt{9 - 4}}{2}$$

$$= \frac{3 \pm \sqrt{5}}{2}.$$

So $a = \dfrac{3 - \sqrt{5}}{2}$ OR $\dfrac{3 + \sqrt{5}}{2}$.

But $\dfrac{3 - \sqrt{5}}{2} < 1$ so $a \neq \dfrac{3 - \sqrt{5}}{2}$

so $a = \dfrac{3 + \sqrt{5}}{2} \approx 2.618$

as we suspected from the plot.